



This most widely used OS is more tolerant than later versions and allows many power

Turn off Task Scheduler

First of all, turn off the Task Scheduler, unless it really is scheduling something important. Most users have no critical programs scheduled, especially after freshly installing Windows. The scheduling utility keeps running in the background, unnecessarily hogging memory. To disable it, double-click the Task Scheduler icon in the system tray and click **Advanced > Stop Using Task Scheduler**. When you create a scheduled task, the feature will be enabled.

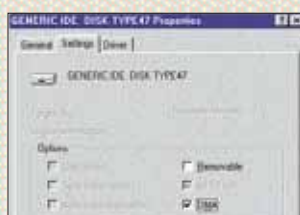
Nix the Active Desktop

For a healthy Windows, you have to get rid of the Active Desktop. Microsoft had added this feature with Internet Explorer 4 and later into Windows when it integrated the browser and the operating system. It allows you to display Web content on the desktop itself. The Active Desktop hogs a ridiculous amount of memory and often causes the system to be unstable. Simply right-click on an empty area of the desktop and disable View as Web Page to turn this off. If you set JPEG or GIF files as your wallpaper, this feature needs to be turned on. It is better to convert them to BMPs before setting them as the wallpaper.

Enable DMA

Direct Memory Access (DMA) is a technique that hard disks and CD-ROM drives can use to transfer data directly to and from memory, without passing through the processor. DMA reduces the load on the system processor since data transfers do not require monitoring by the CPU. With DMA, a write or read operation can be executed in two to four clock cycles. Without DMA it will cost the CPU a minimum of 16 clock cycles per operation. Not only do disk read/write operations gain a significant boost in speed, but Windows also works faster, since it can load system files much faster and also accesses the swap file at higher speeds.

Older hard disks and CD-ROM drives do not support DMA, but almost all computers today use DMA-compliant devices. In case you face a problem after enabling DMA, reboot Windows to Safe Mode and disable the option. Also, DMA needs to be enabled from the BIOS,



Enabling DMA significantly improves speed

but again, on most computers, this is on by default. DMA is supported only in Windows 95 OSR 2 and later.

To enable DMA, open **Control Panel > System > Device Manager**. Expand the Disk drives tree, select the hard disk and click Properties. Under the Settings tab, check the DMA option. Repeat this for all other hard disks and the CD-ROM or DVD-ROM drives.

File system properties

Changing the File System settings can also provide better performance. First, change your computer's role to a Network Server, even if it is not one. Open **Control Panel > System > Performance** and click File System. Under the Hard Disk tab, change the typical role of the computer to Network server instead of Desktop computer. The setting controls the size of various internal data structures used by the 32-bit file access driver (VFAT). When you use the Desktop computer setting, VFAT allocates memory to record the 32 most recently accessed folders and 677 most recently accessed files, consuming approximately 10 KB of memory. With the Mobile or docking system setting, VFAT allocates memory to record 16 folders and 337 files, taking up around 5 KB of memory. As a Network server, VFAT allocates around 40 KB of memory to record 64 folders and 2,729 files. With the kind of memory available on sys-

tems today, this slight overhead hardly makes any difference, especially considering the performance gain it provides.

Optimise virtual memory

The way virtual memory is configured tremendously affects Windows performance. This is true for all versions of Windows, including the now-obsolete Windows 3.1. The swap file is what Windows uses to store temporary data when it runs out of RAM. Thus, your working is not limited by the amount of RAM on your machine. However, for a program to be able to use this data, it must be transferred back to the RAM. The data that is not immediately required is moved to a part of the hard disk and recalled when required. However, since hard disks are nowhere as fast as RAM, swapping data back and forth drastically reduces speed. No matter how much RAM you have, Windows will always use the swap file for some infrequently used parts of the OS (read the next tip to see how to avoid this).

By default, Windows



Optimise the Windows swap file for better performance